1A, 20V - 150V Schottky Barrier Surface Mount Rectifier

FEATURES

- Ideal for automated placement
- Compact package size, profile <0.85mm
- High surge current capability
- Low power loss, high efficiency
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free according to IEC 61249-2-21

APPLICATIONS

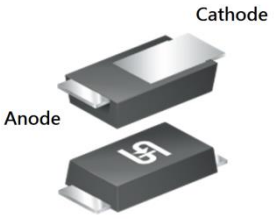
- Switching mode power supply (SMPS)
- Adapters
- DC to DC converter

MECHANICAL DATA

- Case: SOD-123HE
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 0.021g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	1	A
V_{RRM}	20 - 150	V
I_{FSM}	30	A
$T_{J\ MAX}$	125, 150	°C
Package	SOD-123HE	
Configuration	Single die	

Pb-Free RoHS COMPLIANT HALOGEN FREE



SOD-123HE

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)								
PARAMETER	**SYMBOL**	**SS12 LS**	**SS13 LS**	**SS14 LS**	**SS16 LS**	**SS110 LS**	**SS115 LS**	**UNIT**
Marking code on the device		12LS	13LS	14LS	16LS	10LS	A5LS	
Repetitive peak reverse voltage	V_{RRM}	20	30	40	60	100	150	V
Reverse voltage, total rms value	$V_{R(RMS)}$	14	21	28	42	70	105	V
Forward current	I_F	1						A
Peak forward surge current, 8.3ms single half sine-wave superimposed on rated load	I_{FSM}	30						A
Junction temperature	T_J	- 55 to +125		- 55 to +150				°C
Storage temperature	T_{STG}	- 55 to +150						°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-case thermal resistance	$R_{\Theta JC}$	25	°C/W
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	70	°C/W

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER		CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage[1]	SS12LS	I_F = 0.5A, T_J = 25°C	V_F	-	-	V
	SS13LS			-	-	V
	SS14LS			-	0.51	V
	SS16LS			-	0.58	V
	SS110LS			-	0.70	V
	SS115LS			-	0.75	V
	SS12LS	I_F = 1.0A, T_J = 25°C		-	0.45	V
	SS13LS			-	0.50	V
	SS14LS			-	0.55	V
	SS16LS			-	0.70	V
	SS110LS			-	0.80	V
	SS115LS			-	0.90	V
Reverse current @ rated V_R[2]	SS12LS SS13LS SS14LS SS16LS	T_J = 25°C	I_R	-	400	μA
	SS110LS SS115LS			-	50	μA
	SS12LS SS13LS SS14LS SS16LS	T_J = 125°C		-	-	mA
	SS110LS SS115LS			-	0.5	mA
Junction capacitance	SS12LS	1MHz, V_R = 4.0V	C_J	80	-	pF
	SS13LS			73	-	pF
	SS14LS			62	-	pF
	SS16LS			48	-	pF
	SS110LS			28	-	pF
	SS115LS			22	-	pF

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE[1]	PACKAGE	PACKING
SS1xLS	SOD-123HE	10,000 / Tape & Reel

Notes:

1. “x” defines voltage from 20V(SS12LS) to 150V(SS115LS)

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

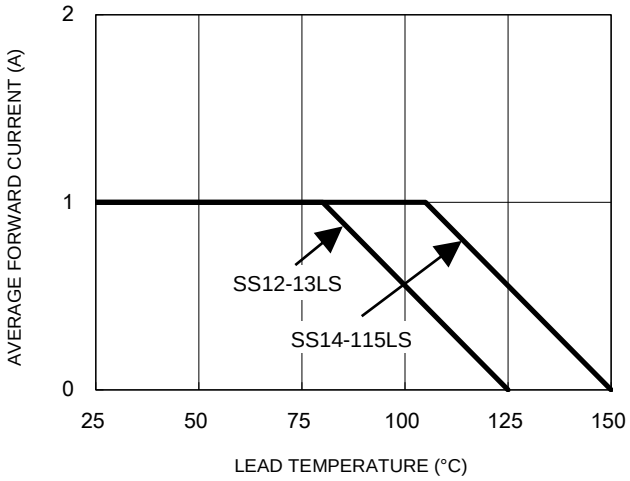


Fig.2 Typical Junction Capacitance

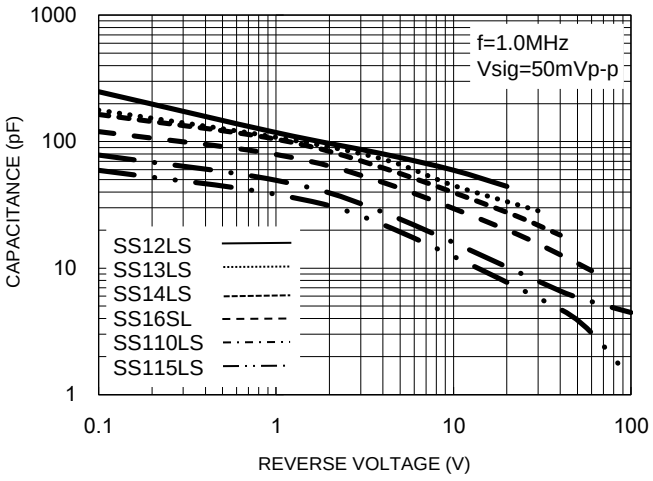


Fig.3 Typical Reverse Characteristics

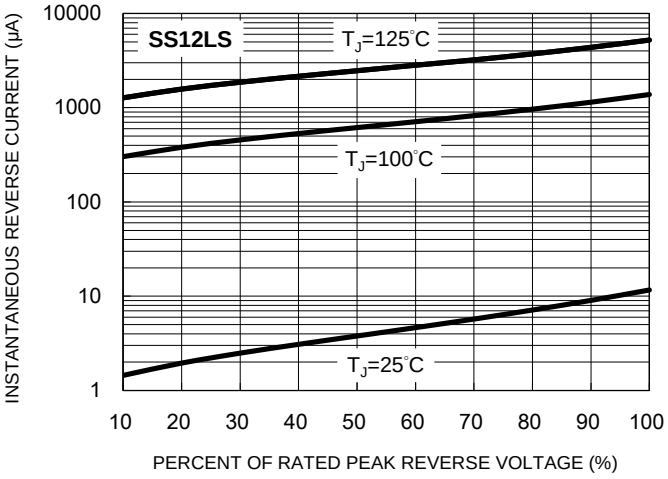


Fig.4 Typical Forward Characteristics

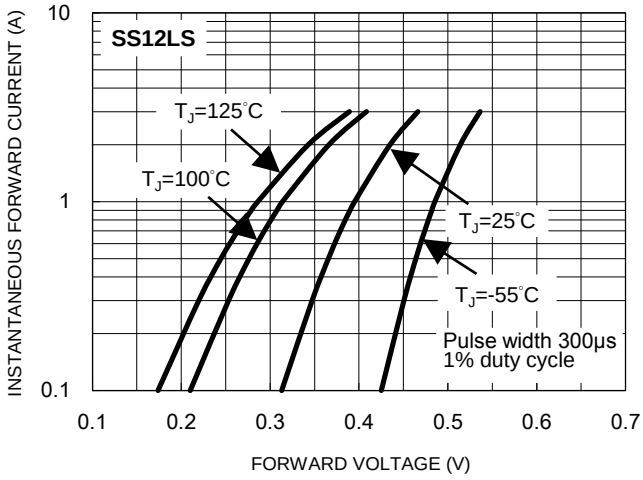


Fig.5 Typical Reverse Characteristics

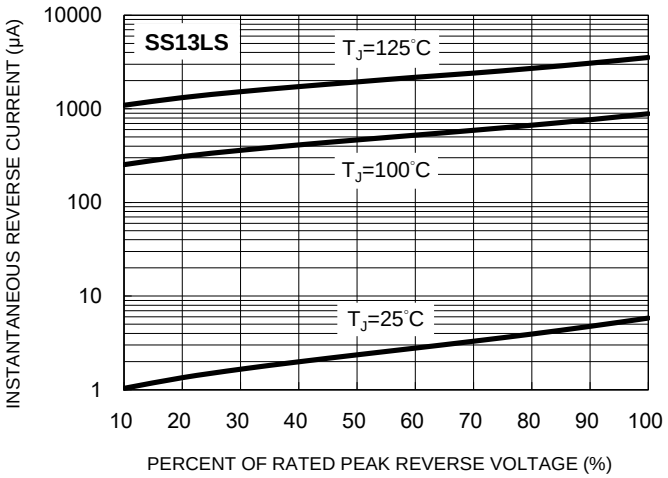
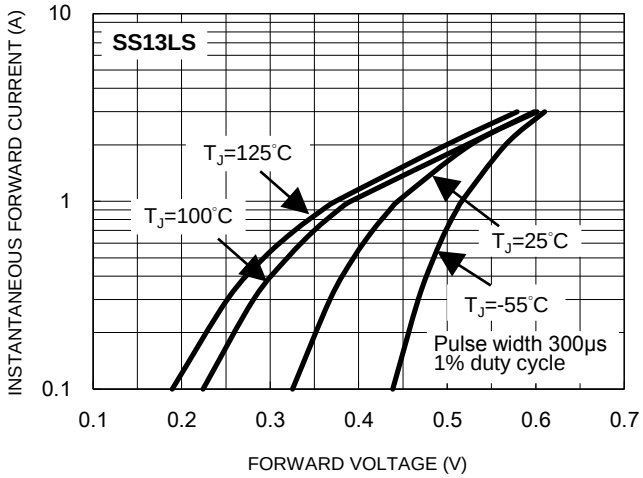


Fig.6 Typical Forward Characteristics



CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.7 Typical Reverse Characteristics

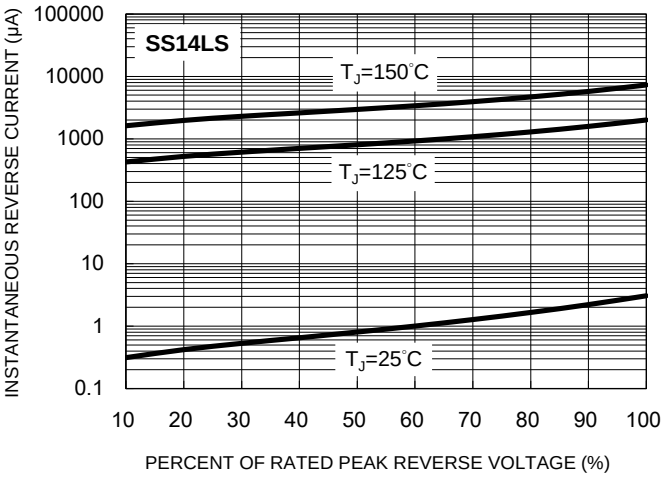


Fig.8 Typical Forward Characteristics

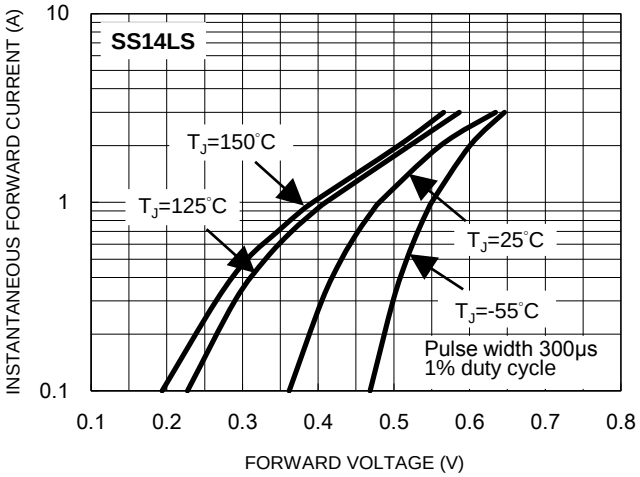


Fig.9 Typical Reverse Characteristics

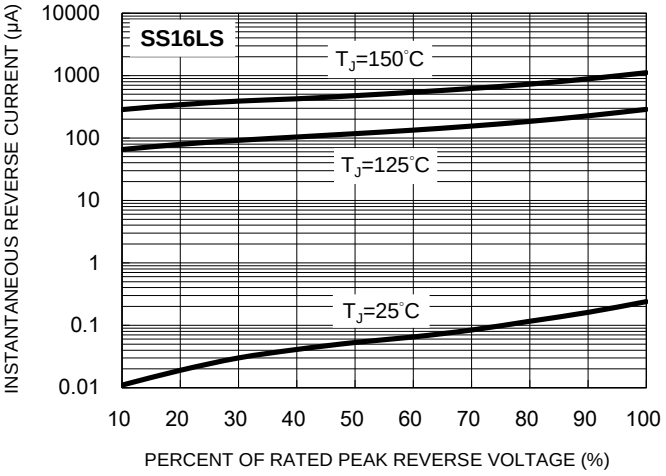


Fig.10 Typical Forward Characteristics

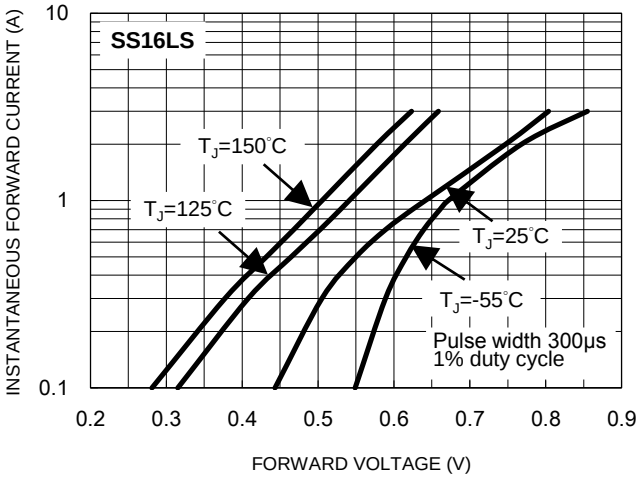


Fig.11 Typical Reverse Characteristics

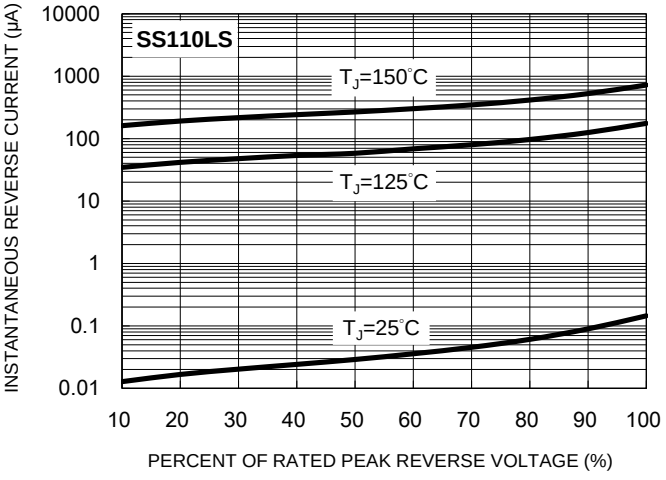
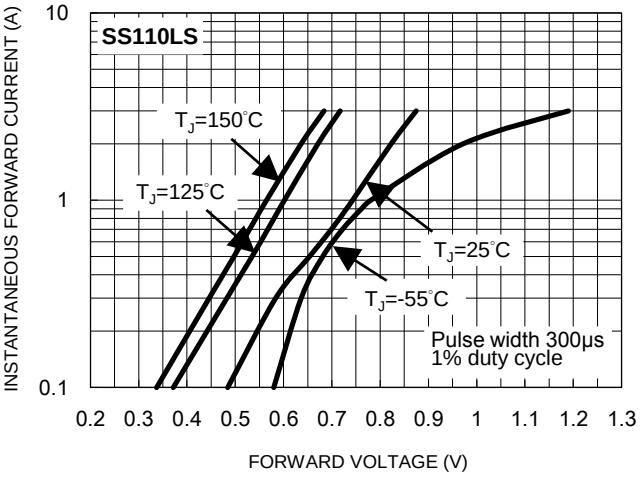


Fig.12 Typical Forward Characteristics



CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.13 Typical Reverse Characteristics

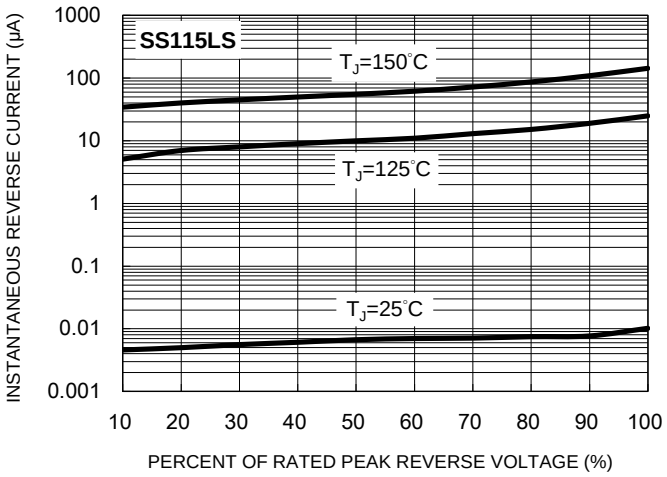


Fig.14 Typical Forward Characteristics

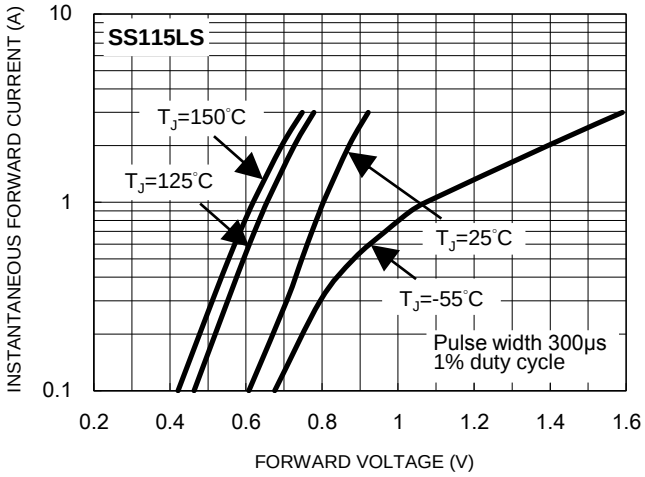
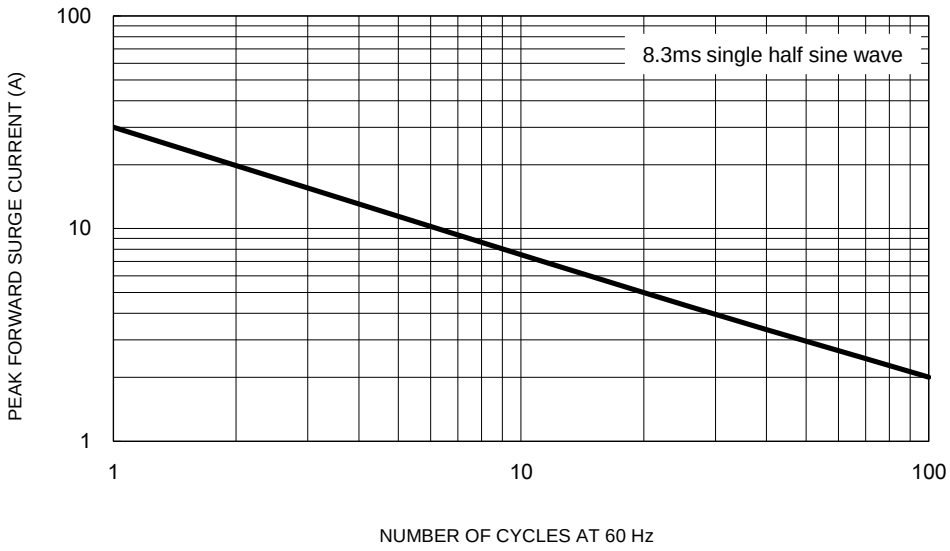
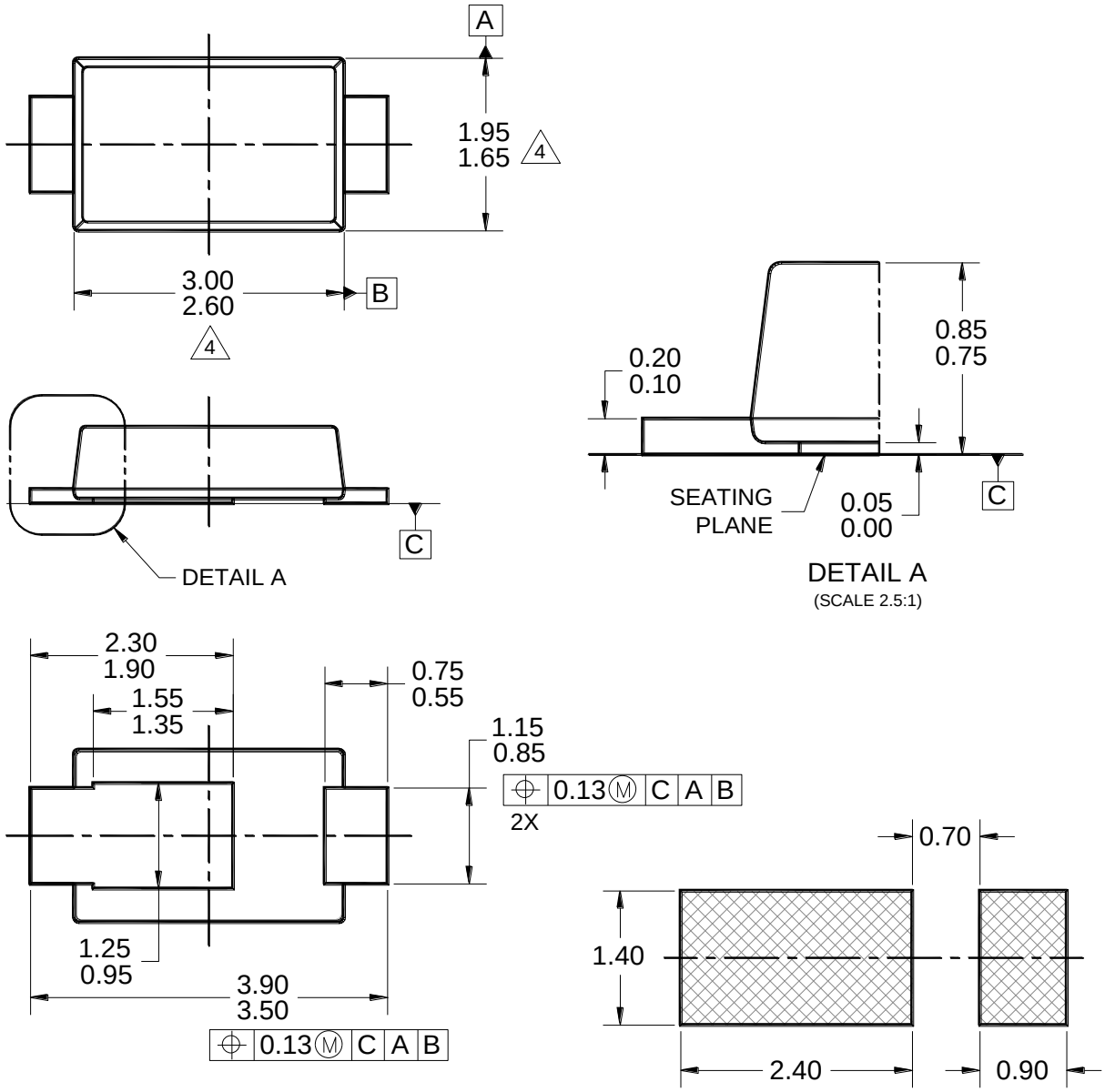


Fig.15 Maximum Non-Repetitive Forward Surge Current

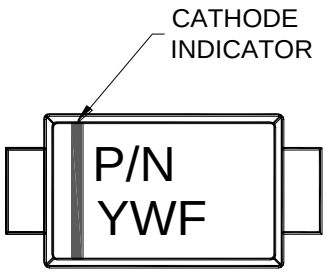


PACKAGE OUTLINE DIMENSIONS

SOD-123HE



SUGGESTED PAD LAYOUT



MARKING DIAGRAM

P/N = MARKING CODE
YW = DATE CODE
F = FACTORY CODE

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-2009.
3. THERE IS NO EXISTING INDUSTRY STANDARD FOR THIS PACKAGE.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH.
5. DWG NO. REF: HQ2SD07-SOD123HE-038 REV A.

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